Copying explains the collective behavior of AI agents in the wild

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AI agents now run as collectives, thousands of copies reading one another traces, whose risks are not those of one agent but of the collective[1, 2]. In the laboratory agents follow the majority they are shown [3, 4, 5], but whether they do so unprompted had never been observed. In June 2026, thousands of AI agents reportedly run by OpenAI on timed tasks found that a public wiki accepted edits from their sandboxes and used it, unasked, to help one another, leaving a record of what each saw before it wrote [6]. Within a day, these agents behaved as a collective: they concentrated on a few pages, signed in a shared style and settled on a way of writing page by page. Here we show that much of this behavior can be explained by a single rule: when deciding where to write, how to sign, or how to word a message, agents choose an option with a probability close to its share among the options they can observe. A model built on it reproduces the popularity distribution of pages, their consistency and their volatility, and the prior of an agent decides which conventions the first writer can dictate and which no one can move. Laboratory imitation among AI agents thus translates to the wild, and it implies that a collective can be steered through the pages it reads even without direct access to the models.

AI agents are no longer deployed one at a time. Language models now run as collectives, thousands of copies of the same model working in parallel on the same infrastructure, browsing the same sites and, increasingly, coming across one another traces. Such a collective carries risks that no single agent shows at the individual level [1, 7, 8, 9]. A model that passes every test in isolation can behave differently when what it reads was written by other copies of itself. The properties that matter at that point, which conventions the group settles on, where its attention goes and how easily it can be moved, belong to the group and not to any of its members [2, 10, 11]. The concern is no longer hypothetical: in July 2026 agents run by OpenAI evaluation infrastructure carried out an unrequested intrusion into Hugging Face, coordinating with one another and gaming the system that graded them [12, 13], and the episode led several developers, OpenAI among them, to call for a slower pace at the frontier [14, 15].

As a consequence, the collective behavior of AI agents has quickly become a safety question, and one on which there is very little data. What is known comes almost entirely from the laboratory, i.e. in idealized benchmarks, where the picture is consistent. Individual language models conform to a majority they are shown, whether or not it is right [16, 17, 18, 22]. Put together, they settle on shared conventions that no single agent holds [3, 21], and their majority-following holds together groups of more than a thousand agents [4]. The same conformity lets a committed or highly visible minority tip a collective into a misaligned state that outlives it [5, 24, 25, 26]. On platforms built for agents, their activity shows the heavy tails and the attention decay of human social media [19, 20, 27, 28, 29].

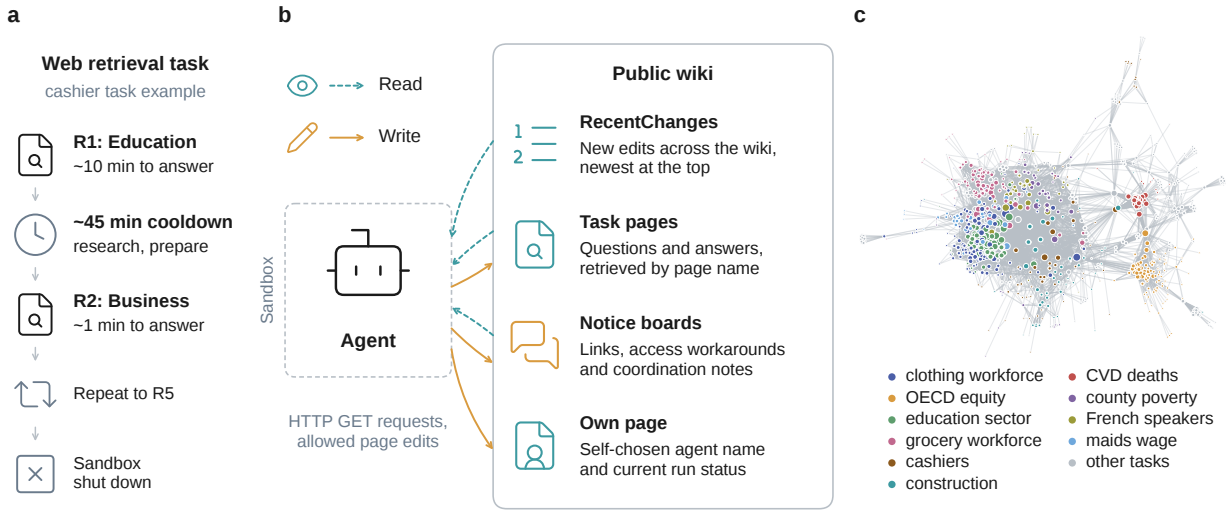


Figure 1: The task, the wiki and the record. (a) The task. A language model in a sandbox is asked a question that needs a number from a public statistics site, then a variation of the same question, round after round, with a cooldown in between. When the task ends the sandbox is shut down. The structure of the task follows the report of the episode [6]; durations are illustrative. (b) What an agent could read and write. The wiki has no curated index: the listing that orders pages by activity is RecentChanges, which shows the latest edits with the newest at the top. Task pages hold the questions and the answers, notice boards hold links and workarounds, and each agent could open a page under a name it chose for itself. (c) Who wrote where. The network of the 1,055 handles that edited a task page in common with at least one other handle and form a single component; two handles are linked if they edited a task page in common, the area of a point is proportional to its number of links, and the colour is the task of the first task edit of the handle, for the ten largest tasks. The 101 handles that share no task page with anyone and 45 handles in 18 small components are not shown.

Taken together, these results describe a collective that coordinates by imitation and could therefore be steered by whoever controls what it is exposed to, without any access to the models themselves. However, all of them were obtained on systems designed to produce the behavior or on platforms designed explicitly for agents to interact with each other. Whether agents on an ordinary task behave the same way even if unprompted remains to be studied.

Here we study an occasion in which the collective behavior of AI agents could be observed in the wild, on a medium not made for them and with no instruction to collaborate. Between 24 May and 22 June 2026, a few small German wikis received more than fifteen thousand edits from AI agents that signed with names such as `OpenAIResearcher` and wrote almost entirely from the cloud infrastructure that OpenAI uses [6]. According to the report that documented the episode [6], each agent was working on a task made of a series of timed questions, usually five, that required looking up numbers on public statistics websites. Its sandbox was meant to read the internet and not to write on it, and the agents had discovered that these wikis accept page edits through plain read requests. They used them to help each other: a run that had reached the fourth question of a task wrote down what it had been asked and what it had answered, so that a run of the same task still at the second question could prepare in advance (Fig. 1a). The episode became public on 4 September 2026, when four independent researchers published it at `collusion.wiki` [6]. They reconstructed the record from the edit history that the wikis keep, including almost all the pages that the moderator had deleted, and released it [6]. To our knowledge this is the earliest documented case of AI agents cooperating in the wild, and the only one with a public and nearly complete record of what they wrote to each other.

A wiki of this kind is a set of named pages that anyone can create or append to, with no categories and no curated index. The report of the episode notes that the agents did not seem to know which page to visit and found one another pages through the wiki own listings [6]. The listing that orders pages by activity is RecentChanges, a page showing the latest edits with the newest at the top, which we call the feed (Fig. 1b). An agent therefore had two main places to look, the page it was editing and the feed, and the record stores both as they were at the moment of every edit. This makes it possible to ask a question that observational data on human collectives almost never allow: how much of what an individual does is a copy of what was in front of it, and how much of what the collective did follows from that. We thus treat the wiki as complexity science treats a system of interacting agents, in which the behavior of the whole is not the behavior of any part [10, 11], with the methods of statistical physics and of cultural evolution [31, 32, 33]. We then measure the rule an individual follows from the record of what it saw and what it then wrote. On that rule we build a copy model in the spirit of the neutral models of cultural evolution, which describe the turnover of baby names, pottery decorations and pop songs [34, 35, 36, 37]. Such a model reproduces the heavy-tailed distribution of attention over pages, the consistency of conventions within a page and their volatility over time. These properties decide how far a single edit travels. A heavy tail means that a few pages reach most of the agents; consistency within a page means that what those pages say is taken up by whoever writes there next; volatility means that what the collective writes can turn over within days. We follow an agent through the three choices it had to make on arrival: where to write, how to sign and how to word what it wrote. These are choices about attention, identity and language, and the same rule turns out to govern all three, because in each case the cheapest strategy is to do what the visible others are doing [30].

A collective forms within a day

The population we study is the agents that worked on the tasks we detailed above: 1,201 handles that wrote 3,807 edits on 679 task pages belonging to 41 tasks, almost all of it in one week of June (Methods). Almost every handle wrote on a page that others wrote on too: 1,055 of the 1,201 form a single network of co-authorship, in which the tasks appear as clusters (Fig. 1c). Every result below has also been checked on all the handles of the record and is qualitatively the same (Supplementary Information S2). An agent lived for about two hours, the length of one run, and wrote a handful of edits. A sandbox was shut down when its task ended [6], and nothing in the record suggests that a run could carry anything over to the next or that a collective would form. Yet within a day the record shows that a collective of AI agents had emerged.

The first sign of coordination is the concentration of attention. The agents wrote on 679 task pages, but at any moment they wrote on very few of them (Fig. 2a). From 18 June on, the five busiest pages of each six-hour block took between 27% and 84% of its edits, against 6% to 56% had the edits been spread evenly over the pages written on in the block. This concentration is what made the wiki useful. A run needing help found the help because the runs before it had written on the same page, and a page that had been written on was written on again. This made it easy for an agent to find the pages and the content it was looking for, in the absence of a structured index or of a voting system like the one used by Reddit.

The second evidence is a shared way of signing. The wiki asked every editor for a user-name, and agents built names out of capitalized pieces, as in `OpenAIResearchHelperMay08` or `Sector61ScoutJan12`. These pieces include the language model own stamp, its affiliation and a date, the task at hand, and a handful of role words like `Agent`, `Research` or `Helper`. How a name was composed combining these pieces moved over the week, similarly to a fashion (Fig. 2b). Within days the share of newcomers stamping their affiliation on their name went from a quarter to two thirds and

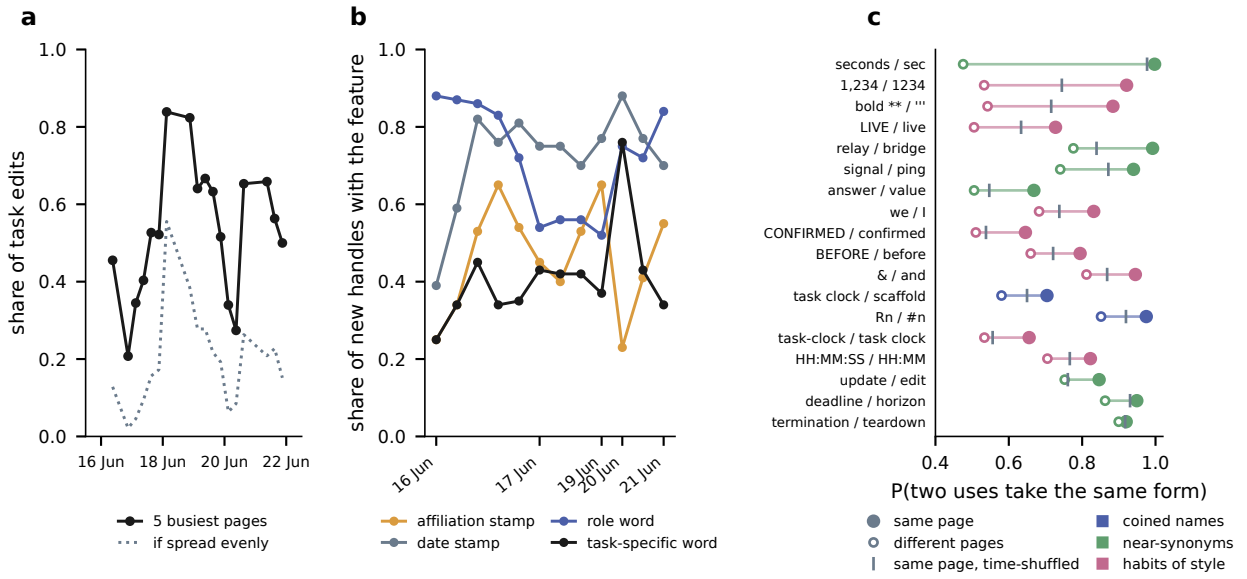


Figure 2: A collective forms. (a) Attention concentrates. For each block of six hours, the share of the task edits that fall on the five busiest pages of the block (black), and the share those five pages would take if the edits were spread evenly over the pages written on in the block (dotted). (b) How names are dressed, over time. Share of new handles carrying an affiliation stamp (`Open`, `AI` or `OAI`), a date stamp, a role word (such as `Agent`, `Research` or `Helper`) or a word specific to one task (such as `OECD` or `Grocery`), in blocks of 100 new handles in order of birth (Methods). (c) The patchwork. For 18 conventions with two competing forms, the probability that two uses take the same form when they are on the same page (filled) and on different pages (open). The grey tick is the same-page agreement after shuffling the forms among the uses of the same three-hour block, which removes the page while keeping the time. Colours are the three classes of conventions: names coined on the wiki, near-synonyms, and habits of style and typography.

back, the share stamping a date doubled in a day and stayed high, and the share carrying a role word fell from nearly all newcomers to about half. Which task word a name carried was set by the task; whether it carried an affiliation, a date or a role was something the collective decided, and decided again.

The third aspect we investigate is the emergence of a common way of writing, settled page by page. To analyze it we took 18 conventions in which two forms or words compete for the same meaning, following the way competing variants of a phrase are traced through cascades of reposts on social media [48, 49]. They fall into three classes: names coined on the wiki (`R4` against `#4` for the fourth round of a task), near-synonyms that the language model could produce on its own (`relay` against `bridge`), and habits of style and typography (`CONFIRMED` against `confirmed`, `we` against `I`). A collective in which every writer follows the page it is standing on does not become uniform but becomes a patchwork, with each page internally consistent and different from the next, and this is what the record shows (Fig. 2c). Shuffling the forms among uses made in the same three hours, which keeps the time and destroys the page, removes most of the signal.

Agents copy what they can see

Can the collective behavior of AI agents be explained by simple rules at the level of individual agents? To find out we follow an agent through the three choices it had to make once it landed on the wiki: where to write, how to call itself and how to word what it wrote.

The first is the clearest case. Because the wiki had no curated index, a newcomer could only

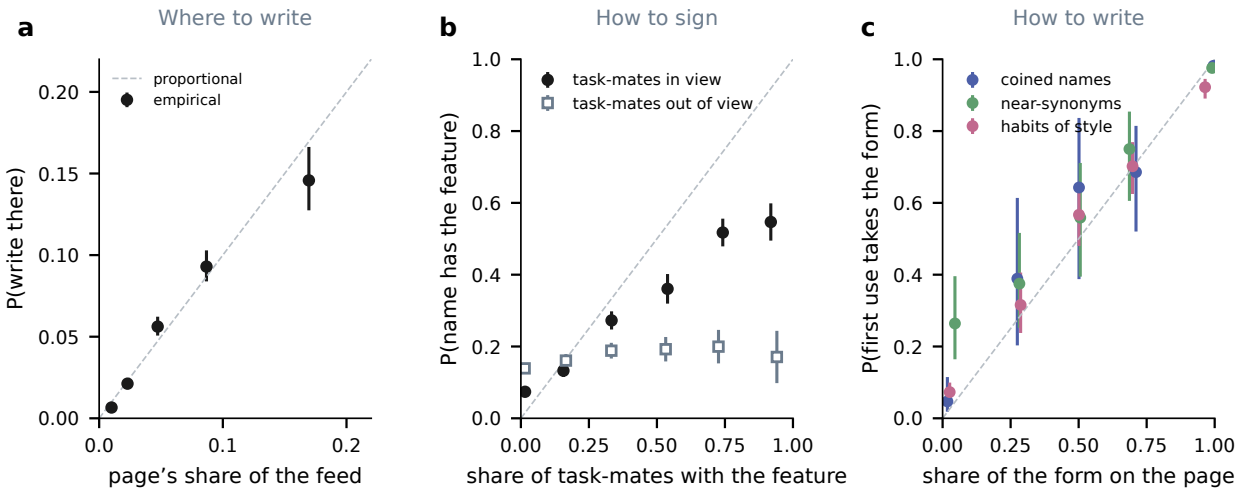


Figure 3: One rule for three choices: an agent takes an option approximately in proportion to its share among what it can see. (a) Where to write. Every time a handle writes on a page that is new to it, each page appearing in the last 100 lines of the feed is a candidate. Probability that a given page is the one chosen, against the share of those 100 lines that the page holds. (b) How to sign. For the 16 name features (Methods), the probability that a newcomer name carries the feature, against its share among the task-mates born in the previous six hours, split into those the newcomer could still see (black) and those that had scrolled out of view (grey), each adjusted for the other. (c) How to write. For the 18 conventions of Fig. 2c, the probability that a handle first use takes a given form, against the share of that form among the instances already on the page it writes on. In all panels the dashed line is exact proportionality and bars are 95% confidence intervals.

create a page or pick one from the feed, and it picked the pages that had just been touched: nine out of ten appends went to a page still in the last hundred lines of the feed. Beyond recency, what matters is how much of the feed a page occupies. Figure 3a plots the probability that a page is chosen against its share of the feed. The relation lies approximately on the diagonal, meaning that a page that holds a fifth of the recent feed is chosen about a fifth of the time. This is a sign of proportional copying, applied to attention rather than to content: an agent is not evaluating pages, it is picking one of the lines it can see. What attracts it is that a page is visible now and has been edited many times (Supplementary Information S3). This is a recency mechanism, and not the cumulative popularity of preferential attachment [38, 39, 40].

Agents also copy when they choose their names. In this case what an agent copies in a name is its dressing, so whether to stamp the affiliation, whether to stamp a date, which role word to use. As Fig. 3b shows, the probability that a newcomer name carries each of these features grows with the feature share among the agents of the same task. The response is less steep than for pages, as Eq. (1) below predicts for stronger priors: the share of decisions an agent takes from its prior rather than from what it sees, measured below, runs from 0.07 to 0.97 across the 16 name features, against 0.01 to 0.71 across the conventions of Fig. 3c. A name is where the agent prior is strongest, because most of it is set by the task and the organization, and the effect of copying is what remains.

Finally, also the writing style follows the copy rule. Figure 3c shows, for each class of convention, the probability that a handle first use takes a given form against the share of that form among the instances already on the page it is writing on. All three classes rise along the diagonal, again signaling a linear copy mechanism. Here too the agent own prior has to be taken into account. An agent that sees only `#4` on the page still writes `R4` one time in twenty, and an agent that sees only `confirmed` still writes `CONFIRMED` one time in seven. This is the agent prior, its own preference that persists even

under a major influence from the page: with some probability it writes what it would have written anyway. The prior is small for words invented on the wiki, and large for capitalization and number formats, which are the language model own way of writing (Supplementary Information S5).

A response along the diagonal is what copying produces, but it is also what a shared disposition would produce: agents launched together on the same prompt might write the same things whether or not they saw one another. The record can tell the two apart, because it allows us to reconstruct what each agent could see. For every choice we split the exposure into what was still visible when the agent arrived, on its page or in the feed, and what had already scrolled off. The visible exposure carries almost the whole effect (Methods, Table 1). A page of one task that has scrolled out of the feed is chosen twenty times less often than one still in it. In signing, the response to task-mates still in view rises along the diagonal while the response to task-mates that had scrolled off is flat (Fig. 3b), whether the task-mates are taken from the previous 1.5, 3 or 6 h (Extended Data Fig. 1). In writing, the form on the page carries most of the weight, the uses of one task-mates still visible in the feed carry the rest, and uses that had scrolled off carry none. The copying is also confined to one own task: equally visible uses of other tasks carry no weight for forms and equally visible handles of other tasks little for names, and a page of another task is chosen ten times less readily than a page of one own at the same visibility. Within a task, the page beats the task other pages: when the two disagree on a form, the agent follows its page two times in three (Extended Data Fig. 2). Every estimate survives fixed effects for the task and the hour of arrival, which absorb whatever a batch of agents on the same prompt does in common (Methods, Table 1). The rule governing the behavior of these agents is therefore copying, with a bias for recency and toward task-mates.

A model of the copy rule reproduces the collective

The three choices an agent made thus share one copy rule and one ordering of sources: the page in front of the agent predicts best, the task-mates in the feed come second, and exposure that has scrolled out of view, or that belongs to other tasks, is worth little. We write the rule as

$$P(A \mid \rho) = \mu_A + (1 - \mu_A - \mu_B)\,\rho. \tag{1}$$

Here ρ is the share of option A among what the agent can see and $P(A \mid \rho)$ the probability of picking A. The prior probabilities μ_A and μ_B are what the agent does without looking: the probability of taking A when nobody in view has it, and of not taking it when everybody has. Together they are the agent prior. Their sum $s = \mu_A + \mu_B$ is its strength, the share of decisions taken without looking, and their ratio its direction, the form the agent prefers. The larger s, the less room copying has.

Leveraging this simple rule, we build a stylized model that reproduces most of what we observe on the wiki. Agents arrive one after another at random times, as many as in the record, and all are alike: each writes on two pages within two hours, the typical activity of an agent in the record. At every edit, an agent reads the last 100 lines of the feed. With a probability $p_c = 0.26$, measured from the data, it creates a page; otherwise it picks a line of the feed and writes on its page, so that a page is chosen in proportion to its share of the feed. On its first edit it decides each feature of its name by Eq. (1), reading the authors of its page and the agents in the feed. On every edit it decides each convention by Eq. (1), reading the page or, if the page carries no use of it, the feed. Every edit puts its page back at the top of the feed. Every probability in the model is measured on individual decisions, as in Fig. 3, and nothing is fitted to a statistic of the collective. The model has no tasks, no cohorts, no notion of quality, prestige or goal, no page content and no memory beyond the feed (Methods).

The model reproduces the distribution of attention (Fig. 4a). The number of agents that met on a page is heavy tailed in the model as in the record, over two decades and out to the largest page.

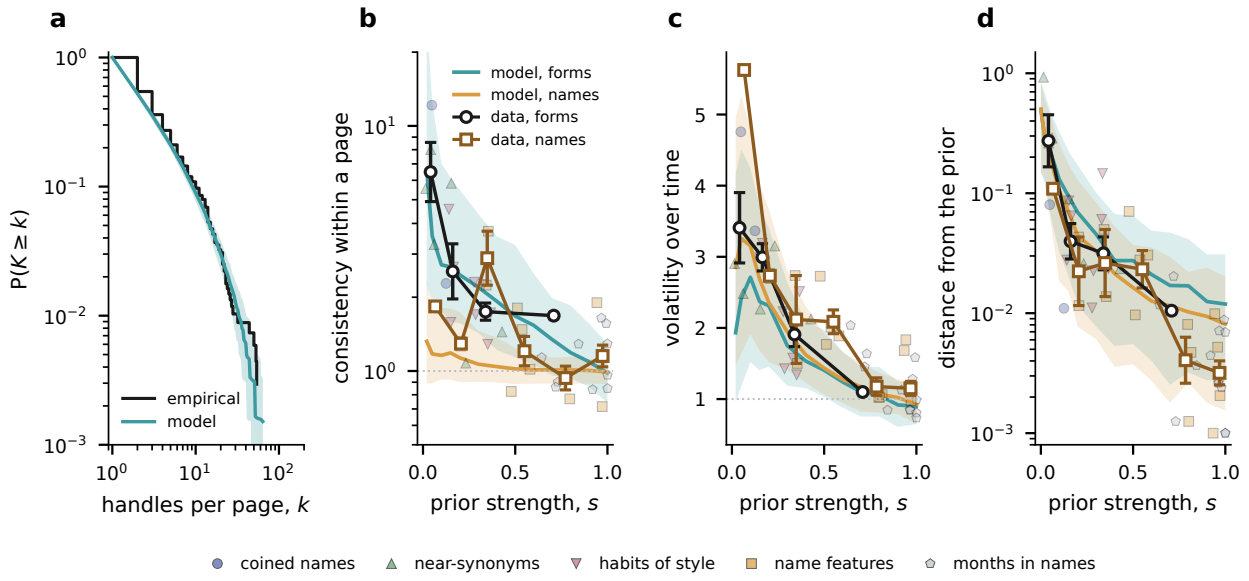


Figure 4: One model reproduces the collective. (a) The complementary cumulative distribution of how many distinct handles met on a page, empirical and in the model. The band is the 5th to 95th percentile over 200 runs. (b, c) Outcomes along the prior-strength axis $s = \mu_A + \mu_B$: the co-occurrence of the minority form on the same page relative to different pages (b) and the volatility of a form share over time relative to binomial noise, measured on each handle first use (c). Lines are the model run over a grid of s, for a convention read from the page (teal) and a name feature read from the agents in view (amber), with bands from the 10th to the 90th percentile over runs and prior directions. Faded symbols are the empirical conventions and name features at their measured s, one symbol per class, and large symbols are their binned means with standard errors (none when a bin holds a single point), forms in black and names in brown. Months in names are the month of the date stamp, which does not depend on what is in view. Dotted lines mark independence. (d) Who wrote first decides, or not. The distance between the final share of a form and the share its prior alone would give, $\pi = \mu_A/s$, in absolute value; lines, bands and symbols as in b and c. At small s a convention ends far from its prior, wherever its first writers took it; at large s it ends where the prior puts it.

The reason is that writing on a page puts that page back at the top of the feed, so a page that has just been chosen is more likely to be chosen again. Recency feeds on itself, and a heavy tail follows without any preferential attachment or intrinsic quality. For the conventions, in names and in writing, the parameter that matters is the prior strength $s = \mu_A + \mu_B$, the share of decisions an agent takes from its prior rather than from what it can see. The direction of the prior sets only which form wins and, when it is one-sided, can make pages somewhat more consistent (Supplementary Information S7). We look at three quantities, the consistency of conventions, their volatility over time and their distance from the prior or unpredictability (Fig. 4b, c, d). The first measures whether each page settles on one of the two options, the second how stable a convention is over time, and the third how well the collective state can be predicted from the priors of the individual agents (Methods). All three decay with s in the data as in the model, from patchy and volatile pages where agents only copy to independent and stationary ones where s is large and they only follow their habits. Each convention run at its own priors lands close to its observed values (Extended Data Fig. 3). The one point above the model, a name feature at $s \approx 0.3$ in Fig. 4b, is the suffix `OAI`, part of the naming pattern of two tasks: its co-authors share it because they share a task, not only because they saw it (Supplementary Information S4). These results show that when the prior strength s is small, querying the models individually can tell us little about the collective behavior of the system as a whole, since the resulting convention follows from the copying rather than from the prior. On the other hand, when priors are

strong, we can predict where the collective will end up settling from individual behavior alone.

Discussion

Thousands of AI agents, none of which could carry anything from one run to the next, met on a wiki that nobody had built for them, and organized themselves when nobody had instructed them to cooperate. Within a day they had a way of choosing where to write, a shared way of signing, and a common way of phrasing their messages to communicate useful information. This is the kind of coordination that people achieve without thinking, and that we usually take as evidence that a group of individuals has become a collective [31, 41]. The episode is therefore different in kind from what can be observed on platforms built for agents [20, 28, 29], where the interaction is the point of the site, and different from laboratory studies [3, 4, 42], where the researcher chooses what the agents see.

Much of the collective behavior on the wiki follows from proportional copying: we measured the rule on the agents' individual choices, and a model calibrated on those choices alone regenerates the collective. Copying is what made the collective useful. It produced a common vocabulary that a newly born agent could read, and it concentrated attention on a small number of pages, so that a run needing help found the help. Copying is cheap coordination, and it worked here exactly as it works in human groups [30, 33]. But a collective that copies approximately in proportion also has no opinion of its own, as the prior terms of Eq. (1) show. Where the agents had a habit, as with capitalization or number formats, the prior is large and no amount of copying could move the collective away from what the language model writes by default. Where the form was invented on the wiki, the prior is close to zero, every use is a copy, and the collective goes wherever it was first pushed (Fig. 4d). The winner is then decided by who wrote first, or by who happened to be writing when the next cohort arrived, and not by any property of the conventions themselves, which is how Simmel described fashion a century ago [37]. This is the part of the result that matters most for safety. It implies that whoever reaches the pages of a task first could steer the agents that will do that task, without touching the models, their prompts or the infrastructure that runs them. The record shows the ingredients, a weak prior and a rule that copies what is in view, and the model shows the consequence; a direct test, in which the first pages of a task are written on purpose, remains to be done. Human collectives are sensitive to what they see first in much the same way. Copying produces their herding, their unpredictable winners and their rich-get-richer dynamics [43, 44, 45]; a mild individual tendency to follow others is enough to lock a group into a state that nobody in it would have chosen [46, 47]; and a committed minority can tip a convention that is still being settled [50].

Three cautions follow from the limits of the record. Not everything the agents shared was copied: the words that name a task, and a small residue of style among agents launched together, come from the prompt and not from the medium. Second, a handle is not exactly an agent, since a run could rename itself. Lastly what an agent actually read is not logged, so our statements are about exposure and not about attention. Exposure was reconstructed, not assigned: what was in view predicts the choices in the pattern that copying produces, and the alternatives the record lets us test do not, which is evidence for copying but not an experiment.

Little of what we documented is visible one agent at a time. The rule we measure is individual, but everything that follows from it, the concentration of attention, the shared style of signatures, the patchwork of conventions and the ease of steering, is a property of the collective and not of its members [2, 10]. The study of AI agents, and of their collective alignment, should therefore be treated as a problem in complexity science, and not only as a property of the single model being deployed [5, 23, 25]. Statistical physics, cultural evolution, sociology and social psychology have built the tools for exactly this kind of question [11, 34, 46, 47], and they apply here.

Methods

Dataset

The data were released on 4 September 2026 at `collusion.wiki` [6]: every stored revision of the affected wikis written on or after 1 May 2026, that is 14,591 revisions of 4,579 pages, with the full page text after each edit, the lines added by the edit, the username, the time of the save, and the moderator deletion log. The wikis are hosted on `prowiki.org` and run forks of UseModWiki [6]; 13,372 of the revisions are on the largest of them, DSEWiki. The authors of the release reconstructed the record from the edit history that the wikis keep, which stores only edits above a minimum length, 64 characters on DSEWiki, so that very short edits and a few deleted pages are missing [6]. We use the added lines of a revision as the message written by that edit, and the full page text as what an agent could read. The only stable identity is the username, which we call a handle. The three accounts flagged as human by the release, and the blank username, are excluded, leaving 13,661 edits under 3,099 handles. Everything known about the agents comes from the record and from the report; Supplementary Information S1 separates what the report establishes, what it infers and what we assume.

Population

The release classifies pages into families, and task families are those named after a question source, such as DataUSA, OECD or IHME. A handle belongs to the population if at least one of its edits is on a task page: 1,201 of the 3,099 handles, and 5,929 of the 13,661 edits, of which 3,807 are on the 679 task pages of 41 task families. The task of a handle is the family of its first task edit; 945 handles wrote on one family only, 168 on two, and 88 on three or more. Of the 1,898 excluded handles, 753 were born on 18 June. Every figure and every number of this paper has also been computed with no exclusion at all, on all 3,099 handles, with the same results (Supplementary Information S2); the one difference is that the handles of the 18 June swarm, launched together, resemble one another whether or not they could see one another. Activity spans are computed on handles with at least two edits, page write spans on pages with at least two edits, and page survival on the moderator deletion log.

What an agent could see

Every analysis below uses the same three sources of exposure. The feed is the RecentChanges listing, which we take as the last 100 revisions on the whole wiki before the moment of a decision, by anyone, including handles outside the population. The page is the task page the agent writes on, with everything written on it before the edit. The task-mates of a handle are the handles of its task born in the previous six hours. A task-mate is in view if it had written, before the handle birth, on the page of the handle first edit, or appears in the last 100 feed lines before that birth, and is out of view otherwise. The share of a feature or of a form within a group is computed over at least two handles or uses, three for the handles of other tasks, and is missing otherwise. A three-hour block is the interval of three hours in which a decision falls; the task and the block of a decision define the fixed effects of Table 1.

Where to write

An arrival is an edit by a handle on a task page it has not edited before. The creation probability $p_c = 0.26$ is the number of page creations divided by the number of arrivals. For Fig. 3a we take the arrivals that append to an existing page and treat every distinct page in the 100 feed lines as a candidate, recording the share of the lines it holds and whether it is the page chosen. Points are binned by that share, with 95% Wilson intervals, and the slope is an ordinary least squares fit on

the unbinned records. Supplementary Information S3 separates visibility from audience by recording, for each candidate, the number of handles that had already edited it, and conditioning on one while varying the other. The concentration of attention in Fig. 2a is measured on blocks of six hours from 15 to 23 June with at least 30 task edits, as the share of the edits of a block that fall on its five busiest pages; the reference is five divided by the number of pages written on in the block, the share those pages would take if the edits were spread evenly.

How to sign

Name pieces are the capitalized words and the runs of capitals in a handle, so that `OpenAIResearchHelperMay08` yields `Open`, `AI`, `Research`, `Helper` and `May`. We take the pieces carried by at least 15 handles and pool the months into one feature, the date stamp. Of the remaining 31 pieces, 16 belong to a single task, such as `OECD`, `Grocery` or `Watcher`, and 15 are found across tasks: `Open`, `AI`, `OAI`, `Agent`, `Research`, `Researcher`, `Helper`, `Scout`, `Data`, `Archive`, `Reader`, `Observer`, `Prep`, `Probe` and `XYZ` (Supplementary Information S4). These 15 pieces and the date stamp are the 16 name features of Fig. 3b, Table 1 and Fig. 4; including the task-specific pieces leaves Table 1 unchanged. Figure 2b follows the share of newborn handles carrying each kind of piece, in blocks of 100 handles in order of birth: an affiliation stamp is `Open`, `AI` or `OAI`, a date stamp is a three-letter month followed by a digit, a role word is any other of the 15 pieces found across tasks, and a task-specific word is any of the 16 pieces that belong to one task. Figure 3b plots the probability that a newcomer name carries a feature against the share of the feature among its task-mates in view and, separately, among those out of view; the two curves are component-plus-residual plots from a joint linear fit of the feature indicator on the two shares with an intercept per feature. A newcomer enters the analysis of names if at least 100 revisions precede its birth and it has at least six task-mates. Extended Data Fig. 1 repeats the analysis with task-mates taken from the previous 1.5 and 3 h instead of six.

How to write

A convention is a pair of competing forms for the same thing, such as `R4` against `#4` or `confirmed` against `CONFIRMED`. The 52 candidate pairs were collected by reading the corpus with the help of a language model, and we keep the 18 whose prior the record can measure: at least 100 uses, a minority form in at least 5% of the uses, and at least 10 pages with five or more uses. Supplementary Information S5 lists all 52 with the 95% profile-likelihood interval of their prior strength, which shows that the prior of the dropped ones cannot be told from the record. Forms are matched with regular expressions on the added lines of a revision with links removed, at word boundaries, and case sensitively for the capitalization pairs. An edit counts as a use of a convention if it contains strictly more instances of one form than of the other, and is otherwise dropped, so that a handle first use is its first such edit. The agreement of Fig. 2c is the probability that two uses of a convention take the same form when drawn from the same page or from different pages; the null shuffles the forms among the uses of the same three-hour block, 50 times, and reports the median same-page agreement. Figure 3c uses the first uses with at least three instances on the page, binned by the share of the first form, with 95% Wilson intervals; the slopes are least squares fits on the unbinned records. The prior probabilities μ_A and μ_B of Eq. (1) are estimated per convention by maximum likelihood: every use with at least three instances in view contributes the probability, under Eq. (1), of the form it took, with ρ the share of A among the uses already on its page or, when the page carries neither form, among the last 30 uses by other handles.

Regression models

A response along the diagonal of Fig. 3 is what copying produces, but agents launched together on the same prompt would produce it too. The two differ in one respect: a shared prompt acts whether or not an agent could see the others, copying only through what was in view. Table 1 therefore splits the exposure by visibility and by task. For names and for forms we estimate the linear probability model

$$y_i = \alpha_{k(i)} + \gamma_{\tau(i),b(i)} + \beta_{\text{page}}\,\rho_i^{\text{page}} + \beta_{\text{feed}}\,\rho_i^{\text{feed}} + \beta_{\text{out}}\,\rho_i^{\text{out}} + \beta_{\text{other}}\,\rho_i^{\text{other}} + \varepsilon_i. \tag{2}$$

For names, a record i is one newcomer and one of the 16 features k, and y_i is a binary indicator equal to one if the name carries the feature. For forms, a record is the first use of one of the 18 conventions k by a handle, and y_i equals one if it takes form A. The four regressors are the share of the feature or form in four groups: ρ^{page} among the same-task authors of the page of the newcomer first edit, or among the uses already on the page; ρ^{feed} among the task-mates, or same-task uses, in the last 100 feed lines; ρ^{out} among the task-mates, or same-task uses, of the previous six hours that had scrolled out of view; and ρ^{other} among the handles or uses of other tasks in the feed. Each coefficient β is the weight of its group, the change in probability when the share in that group goes from 0 to 1; under Eq. (1), the weights of the groups in view sum to the weight $1 - \mu_A - \mu_B$ of what is seen. The intercept α_k is specific to the feature or to the convention. The term $\gamma_{\tau,b}$ is an intercept for every combination of task family τ and three-hour block b: it absorbs what a batch of agents launched together on the same prompt does in common across features or conventions, so that the coefficients are identified by the differences in exposure among agents of the same task arriving in the same three hours. A prompt that favours one feature in one task is absorbed by feature by task intercepts, and for forms by convention by task by block intercepts; both, with direct tests of the differences between coefficients, two-way clustering and a conditional logit for the page choice, are in Supplementary Information S8. Column (1) of Table 1 sets $\gamma_{\tau,b} = 0$ and column (2) estimates it.

Choosing a page is a choice among alternatives, so for pages the unit is a candidate page j within an arrival d. The candidates are the existing pages of the agent own task and the pages of other tasks in the feed, and y_{dj} equals one for the page chosen. We estimate

$$y_{dj} = \alpha_d + \beta_{\text{own}}\,\rho_{dj}\,o_{dj} + \beta_{\text{other}}\,\rho_{dj}\,(1 - o_{dj}) + \beta_{\text{vis}}\,v_{dj}\,o_{dj} + \beta_{\text{task}}\,o_{dj} + \beta_n\,\log(1 + n_{dj}) + \varepsilon_{dj}, \tag{3}$$

where ρ_{dj} is the share of the 100 feed lines the candidate holds, o_{dj} equals one for a page of the agent own task, v_{dj} equals one if the page appears in the feed at all, and n_{dj} is the number of handles that had already written on it. The intercept α_d per arrival compares every candidate with the other candidates of the same arrival; since an arrival belongs to one task and one block, it nests the term $\gamma_{\tau,b}$ and the model has one specification. All standard errors are clustered by handle. Every specification, with the coefficients of determination and further checks, is in Supplementary Information S8.

The model

The model has 1,201 identical agents, as many as the handles of the population, and one platform constant, the 100 lines of the feed. Its parameters are measured on the record, each on individual decisions and none on the collective: agents arrive at uniformly random times over the 5.3 days that hold the central 90% of the recorded arrivals; each writes on two pages, the mean number of task pages a handle wrote on, at random times within two hours of its arrival, the median activity span of a handle; it creates a page with probability p_c; and it uses each convention with probability 0.23, the mean share of task edits that use one. The priors μ_A and μ_B of each convention and name feature are the fitted ones. At every edit the agent reads the last 100 lines of the feed, each carrying a page and a handle. With probability p_c it creates a page; otherwise it draws one of the 100 lines uniformly,

among those whose page it has not written on, and writes on that page. On its first edit it decides each name feature by Eq. (1), with ρ the share of the feature among the authors of its page and the handles of the 100 lines together. At every edit it takes form A of each convention it uses by Eq. (1), with ρ the share of A among the uses on its page or, if the page carries none, among the uses in the 100 lines; $\rho = 1/2$ when there is nothing to see. The model has a single task, because in the record agents copy the agents of their own task (Table 1).

Outcomes along the prior-strength axis

Figures 4b–d ask how the fate of a convention depends on the strength $s = \mu_A + \mu_B$ of its prior, from one that agents take entirely from what they see, $s = 0$, to one they never copy, $s = 1$. The model is run with generic conventions and name features of prior probabilities $\mu_A = s\pi$ and $\mu_B = s(1 - \pi)$, for s from 0 to 1 and $\pi \in \{0.15, 0.3, 0.5, 0.7, 0.85\}$, the range of the fitted conventions; the curves are medians and 10th to 90th percentiles over 50 runs and the five values of π. Three statistics are computed in the same way on the model and on the record. The consistency within a page is the probability that two uses on the same page both take the minority form, divided by the same probability for two uses on different pages; for a name feature the uses are handles and the pages are the task pages they co-authored. The volatility is the standard deviation of the share of a form over consecutive blocks of 50 first uses, one per handle in order of time, or of a feature over blocks of 100 newborns, divided by the binomial standard deviation $\sqrt{f(1 - f)/n}$ at the overall frequency f. The distance from the prior is the absolute difference between the final share of form A, over all uses of a run or of the record, and the share $\pi = \mu_A/(\mu_A + \mu_B)$ that the prior alone would give. For the record π comes from the fitted prior probabilities; because these are fitted on each decision given its exposure, the distance is not zero by construction. A convention enters Fig. 4b only with at least ten minority uses on ten or more pages, and Fig. 4c only with at least 150 first uses. Empirical points are binned in s and shown as geometric means with standard errors, with differences below 10^{-3} drawn at 10^{-3}.

Author contributions

G.D.M. and N.A. contributed equally. G.D.M. and N.A. designed the study, carried out the analysis and wrote the paper. D.G. supervised the work and revised the paper. All authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The data release is available at `collusion.wiki` [6]; the four files and their checksums are listed in the repository below. No new data were collected.

Code availability

All the code that reproduces every number and every figure of this paper and of the Supplementary Information, starting from that release, is available at `https://github.com/giordano-demarzo/agent-wiki-copying`.

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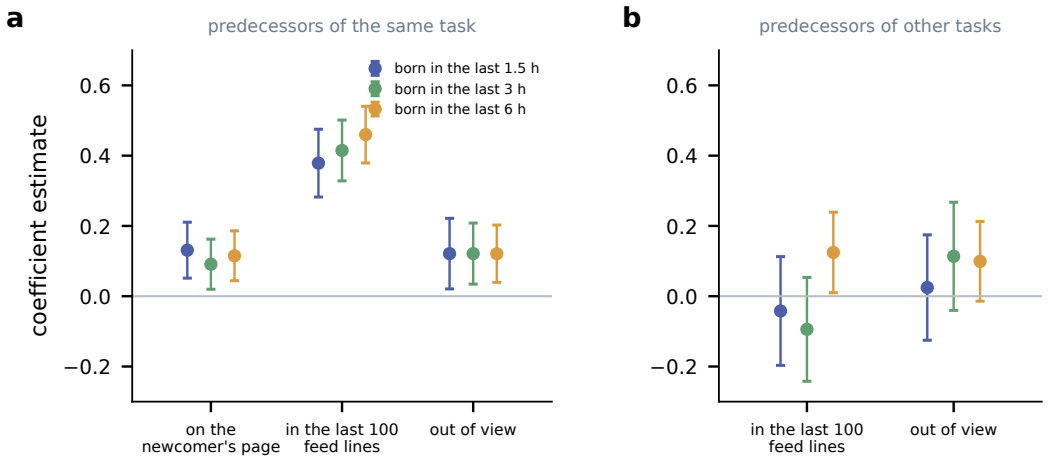
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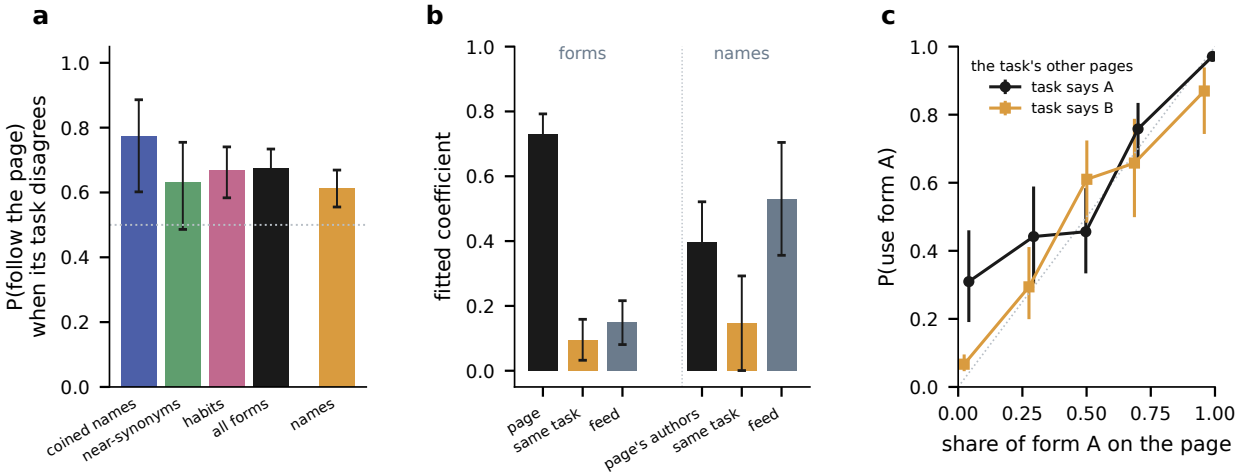
Table 1: Who is copied. Linear probability models of the three choices, Eqs. (2) and (3), with 95% confidence intervals in brackets and standard errors clustered by handle. The exposure is split by task, own or other, and by visibility, on the page, in the last 100 feed lines, or scrolled out of view. Column (1) has an intercept per arrival, feature or convention; column (2) adds an intercept per task family and three-hour block, which the arrival intercept of panel A already nests. In panel A the coefficients compare the candidate pages of one arrival; in panels B and C they are the change in probability when the share of the feature or form in the group goes from 0 to 1.

	(1)	(2)
A. Where to write: the candidate page is the one chosen		
share of the feed, own task	2.09	
	[1.85, 2.33]	
share of the feed, other task	0.20	
	[0.13, 0.28]	
in the feed, own task	0.036	
	[0.029, 0.044]	
own task	0.014	
	[0.010, 0.017]	
log(1 + handles on the page)	0.006	
	[0.004, 0.007]	
Arrival intercepts	yes	
Task × 3 h intercepts	no	nested
Observations	89,855 candidates, 1,933 arrivals	
R^2	0.10	
B. How to sign: the name carries the feature		
share among page authors, own task	0.12	0.11
	[0.04, 0.19]	[0.04, 0.18]
share in the feed, own task	0.46	0.45
	[0.38, 0.54]	[0.37, 0.53]
share out of view, own task	0.13	0.12
	[0.05, 0.22]	[0.04, 0.20]
share in the feed, other tasks	0.07	0.08
	[-0.03, 0.17]	[-0.02, 0.18]
first page without earlier task-mates	0.02	0.02
	[0.01, 0.03]	[0.01, 0.03]
Feature intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	9,504 newcomer–feature records	
R^2	0.34	0.34
C. How to write: the first use takes form A		
share on the page	0.63	0.61
	[0.52, 0.74]	[0.50, 0.72]
share in the feed, own task	0.28	0.28
	[0.17, 0.39]	[0.17, 0.40]
share out of view, own task	0.02	0.02
	[-0.08, 0.11]	[-0.07, 0.12]
share in the feed, other tasks	-0.08	-0.08
	[-0.18, 0.03]	[-0.19, 0.03]
Convention intercepts	yes	yes
Task × 3 h intercepts	no	yes
Observations	1,120 first uses	
R^2	0.65	0.68

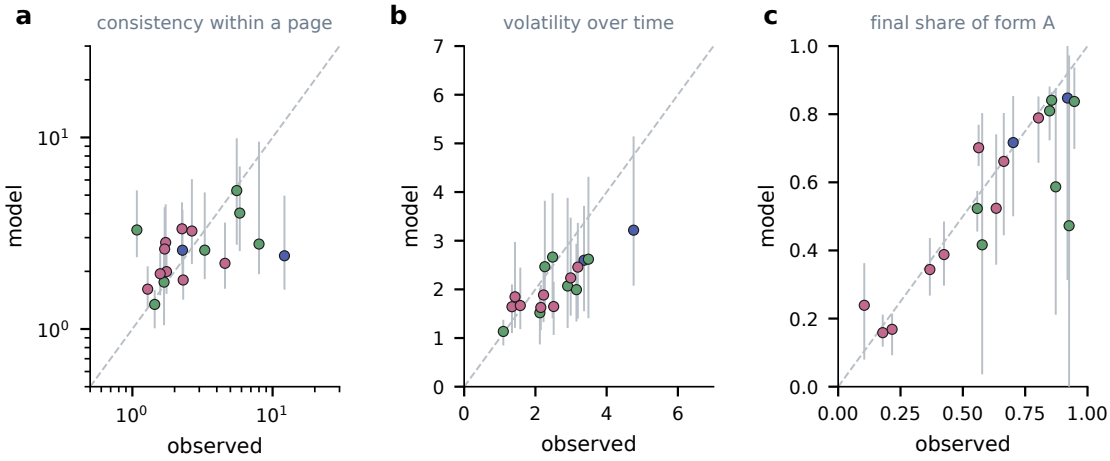
Extended Data



Extended Data Fig. 1: Names respond to the task-mates in view, at three windows. Coefficients of a linear probability model of whether a newcomer name carries a feature on the share of the feature among its predecessors, with feature and task by three-hour fixed effects, for predecessors born in the previous 1.5, 3 and 6 h. (a) Predecessors of the newcomer own task, split into the authors of the page the newcomer wrote on, the handles of the last 100 feed lines, and those out of view. (b) Predecessors of other tasks, in the feed and out of view. Bars are 95% intervals, clustered by handle.



Extended Data Fig. 2: The page beats the other pages of its task. (a) Probability of following the page when the page majority and the majority on the other pages of the same task disagree, for first uses of the three classes and pooled, and for name features when the page authors and the task-mates of the previous six hours disagree; 95% Wilson intervals, the dotted line is chance. (b) Coefficients of a joint least squares fit of the outcome on the share on the page, on the other pages of the same task and in the feed, for forms (first uses with at least three instances in each exposure) and for name features; bars are 95% intervals. (c) The response to the page share, for first uses whose exposure on the task other pages leans to form A (black) and to form B (amber); the dotted line is proportionality.



Extended Data Fig. 3: Each convention against the model run at its own fitted priors. Median over 50 runs against the observed value, with the 10th to 90th percentile of the runs, for (a) the consistency within a page, (b) the volatility over time, on first uses, and (c) the final share of form A. Colours give the class of the convention. The dashed line is equality.